

CHAKRADHAR REDDY PEDDAVENKATAGARI

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TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, C++, C
- **Web & Backend:** HTML, CSS, REST APIs, Streamlit, API Integration
- **AI / ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, LLMs, RAG, NLP, Computer Vision
- **Data & Libraries:** NumPy, Pandas, FAISS, LangChain
- **Tools & Cloud:** Git, GitHub, AWS, Firebase, Jupyter Notebook, Google Colab

EXPERIENCE

Research Assistant – AI Software Development

Jan 2026 – Present

University at Buffalo SUNY

- Developed a modular assistive-navigation pipeline using **Python and PyTorch**, integrating a frozen DINOv2 vision encoder, semantic segmentation decoder, and task-specific output components.
- Built reusable modules for ADE20K data loading, preprocessing, training, validation, checkpointing, and inference, enabling reproducible experimentation and easier system maintenance.
- Optimized model training and evaluation using mixed-precision execution, automated checkpoint selection, and quantitative monitoring, achieving **50.1% validation mIoU** and **83.2% pixel accuracy**.
- Designed extensible interfaces for future intent-router and vision-language integration while documenting findings that contributed to a competitively selected abstract presented at the **Stony Brook AI Symposium**.

Software Development Intern

Jan 2025 – Jun 2025

Excl Solutions Pvt. Ltd.

- Developed and optimized end-to-end **Python-based image-processing pipelines** for large-scale TEM image analysis, improving processing throughput by up to **80%**.
- Built an embedding-based semantic retrieval application that enabled natural-language search across internal datasets and documents, improving access to relevant technical information.
- Implemented modular components for data ingestion, preprocessing, model inference, vector retrieval, and result generation, improving code maintainability and workflow reusability.
- Collaborated with technical stakeholders to translate requirements into software improvements, troubleshoot pipeline issues, and deliver reliable functionality, earning a **Letter of Recommendation from the CEO**.

PROJECTS

RAG-Based Document Question Answering System (LLM + Vector Search)

github.com/RAG_Chatbot

- Built an end-to-end **RAG-based document question-answering system** to reduce unsupported LLM responses by grounding generated answers in content retrieved directly from uploaded PDF documents.
- Designed the retrieval pipeline using **LangChain, FAISS, GoogleGenerativeAIEmbeddings, and Gemini API**, splitting documents into 800-character chunks with 150-character overlap and implementing vector indexing and similarity search.
- Developed a **Streamlit** interface supporting multi-document question answering, successfully testing retrieval across **3 PDFs simultaneously** and observing more relevant, document-grounded responses compared with the initial document-retrieval approach.

Robust Image Classification under Spurious Correlations

github.com/spurious-correlation-classification

- Built a robust image-classification pipeline on the **Waterbirds** dataset to reduce background-driven shortcut learning, using **Grad-CAM, PCA, and worst-group accuracy** to diagnose model bias.
- Implemented and compared frozen **ViT** and **DINOv2** feature extractors with **ERM, Core-Aligned Regularization, and Group DRO**, selecting DINOv2 after observing cleaner feature separation in PCA.
- Achieved the strongest performance with **DINOv2 + Group DRO**, reaching **97.6% accuracy, 94.8% F1, and 95.3% worst-group accuracy**, compared with **66.8% worst-group accuracy** using ViT features + ERM.

PUBLICATIONS

Neural Sequence-to-Sequence Modeling with Attention for Abstractive Text Summarization

Apr 2024

- Designed a neural sequence-to-sequence model with attention for abstractive text summarization. This approach enhances contextual understanding, improving the accuracy of generated summaries.
View Paper: arxiv.org/abs/2404.08685

EDUCATION

Master of Science in Computer Science and Engineering

Aug 2025 – May 2027

University at Buffalo SUNY, New York

CGPA: 3.958 / 4

Bachelor of Technology in Computer Science with Specialization in Cloud Computing

Jun 2021 – May 2025

SRM University, Chennai, India

CGPA: 9.64 / 10

ACCOMPLISHMENTS

- Merit-Based Scholarships totaling ₹1.2L (2022–2024)
- Ranked 1st in class and 2nd globally in a Kaggle-hosted Deep Learning wildfire prediction competition
- Supported student event management, strengthening leadership, teamwork, and coordination skills
- Demonstrated leadership as a Student Leader in campus dining operations, managing 20+ student employees and coordinating daily workflows